

Certificate



No.: V 37.09/14

Product tested	Electromagnetic Valves with Safety Function	Certificate holder	Norgren GmbH Stuttgarter Str. 120 70736 Fellbach Germany
Type designation	24011, 24010		
Codes and standards	IEC 61508 Parts 1-2 and 4-7:2010	IEC 61511 Parts 1-3:2004	
Intended application	Safety function: Safe return to 0-position The achieved failure rates enable the usage in safety-related systems up to SIL 3 acc. IEC 61508. Constraints regarding the necessary Hardware Fault Tolerance (HFT) from the relevant application standard (e.g. IEC 61511) have to be considered.		
Specific requirements	The instructions of the associated Installation and Operating Manual shall be considered. Summary of test results see backside of this certificate.		
Valid until	2019-11-17		

The issue of this certificate is based upon an examination, whose results are documented in Report No. V 37.09/14 dated 2014-11-17.
This certificate is valid only for products which are identical with the product tested. It becomes invalid at any change of the codes and standards forming the basis of testing for the intended application.

TÜV Rheinland Industrie Service GmbH
Bereich Automation
Funktionale Sicherheit
Am Grauen Stein, 51105 Köln

Köln, 2014-11-17

Certification Body for FS-Products


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Manufacturer Norgren GmbH
 Stuttgarter Straße 120
 70736 Fellbach, GERMANY

Product tested Type series 24011 / 24010

Device-Specific Values:

		24011 / 24010
Probability of Failure on Demand	p (failure/Demand)	2 E-07
Confidence Level	$1-\alpha$	95 %
Safe Failure Fraction	SFF	< 99 %
Hardware Fault Tolerance	HFT	0
Diagnostic Coverage	DC	0
Type of Subsystem	IEC 61508-2, 7.4.3.1.2	Typ A

Derived Values for Low Demand Mode (LDM)

Example with $f_{np} = 1$ / Year		24011 / 24010	
Assumed Demands per Year	f_{np}	1,14 E-04 /h	1 /y
Assumed Test Intervall *	T_i [y]	8760 h	1 y
Dangerous Failure Rate	$\lambda_D = p \times f_{np}$	2,28 E-11 /h	0,02 FIT
MTBF dangerous Failures	$MTBF_D = 1/\lambda_D$	4,39 E+10 h	5.000.000 y
Safe Failure Rate	λ_S	2,26 E-09 /h	2 FIT
Total Failure Rate	$\lambda_S + \lambda_D$	2,28 E-09 /h	2 FIT
$MTBF_{total}$	$1/(\lambda_S + \lambda_D)$	4,38 E+08 h	50.000 y
dangerous detected	λ_{DD}	0 /h	0 FIT
dangerous undetected	λ_{DU}	2,28 E-11 /h	0,02 FIT
safe detected	λ_{SD}	0 /h	0 FIT
safe undetected	λ_{SU}	2,26 E-09 /h	2 FIT
Average Probability of Failure on Demand	$PFD_{avg,1001}$	1 E-07	

* Test Interval can be extended in responsibility of the end user under consideration of ambient conditions and safety relevant parameters.

Derived Values for High Demand Mode (HDM)

Example with $f_{np} = 2000$ / Year		24011 / 24010	
Assumed Demands	f_{np}	0,23 /h	2000 /y
Dangerous Failure Rate	$\lambda_D = p \times f_{np}$	4,6 E-08 /h	46 FIT
MTBF dangerous Failures	$MTBF_D = 1/\lambda_D$	2,17 E+07 h	2174 y
Safe Failure Rate	λ_S	4,55 E-06 /h	4554 FIT
Total Failure Rate	$\lambda_S + \lambda_D$	4,6 E-06 /h	4600 FIT
$MTBF_{total}$	$1/(\lambda_S + \lambda_D)$	2,17 E+06 h	248 y
dangerous detected	λ_{DD}	0 /h	0 FIT
dangerous undetected	λ_{DU}	4,6 E-08 /h	46 FIT
safe detected	λ_{SD}	0 /h	0 FIT
safe undetected	λ_{SU}	4,6 E-06 /h	4600 FIT

Source of Derived Values

The named values are the result of extensive laboratory test. Reliability of safety function was proved under harsh conditions. Furthermore test results have been assured by proven-in-use data of the last five years. Random and systematic failure which are caused by the manufacturer have been taken into account.

Time of Usage

A time of usage of more than 12 years for BR 24011 and 6 years for BR 24010 (+ 2 years of storage) can only be favored under responsibility of the operator, consideration of specific external conditions (securing of required quality of control air, max. temperature, time of impact), and adequate test cycles.

Quality Management

These statements are bound to a proven and verified deployment of safety-related quality management of the manufacturer.